The Open Source Firefighter

A specification for coordinated neighbourhood ember defense

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> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.**

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This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

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Preface

This document specifies the Open Source Firefighter, a coordinated defensive infrastructure system designed to protect residential structures, federal facilities, and remote installations during the ember-cast window of wildland-urban interface fires. The specification defines capability requirements, example implementations, coordination protocols, integration interfaces, and operational logging sufficient for competent operators or contractors to build compliant installations across the range of deployment environments the actual problem requires.

The specification is published under permissive open licence. Implementations may be built by homeowners, by contractors, by federal land management agencies, or by manufacturers. Substitution of components meeting the capability requirements is expressly permitted and expected. This is a family of systems defined by function rather than by specific product.

This specification is the spine of a document family. Hardware detail lives in Annex A; wire format and vocabulary in Appendix C of this document; a known-good reference configuration in the parts catalogue at Annex A, Appendix A; employment doctrine in the Tactics Paper.

This specification incorporates in full: the Layered Architecture, the Event Interface and Message Category Enumeration, Backhaul and Gateway Nodes, Expeditionary Deployment, and the Governance Charter, each at the location its original front matter directed.

The Problem Addressed

Ember cast, not radiant heat, destroys most structures in wildland-urban interface fires. Embers travel ahead of the flame front and land on roofs, in gutters, in dry vegetation adjacent to structures, and in other ignitable receiving sites. Small ignitions grow into structural fires while the flame front is still distant. Keeping receiving surfaces wet during the ember-cast window prevents these ignitions.

The defensive principle at the individual property level is well understood. Sprinkler and cannon systems that wet roof, gutter, and perimeter surfaces during fire events reduce structure loss substantially. The engineering has been demonstrated. What has not been demonstrated at scale is coordination of individual property systems across whole neighbourhoods sharing a common water supply, or integration of residential systems with federal facility and remote installation defenses in mixed-jurisdiction fire environments.

This specification addresses the coordination problem across the range of participants that actual wildland-urban interface events involve. Multiple houses on a common water supply cannot simultaneously operate at maximum draw without exhausting the supply. Federal facilities surrounded by residential development benefit from neighbourhood-scale coordination but require appropriate handling of their internal operational data. Remote installations on federal land require radio infrastructure appropriate to their geographic scale and reliability requirements. Professional incident command arriving on scene requires unified operational picture across all participating systems regardless of the underlying radio infrastructure.

System Architecture

Overview

Each participating installation has an autonomous local system consisting of a water supply (municipal connection, cistern, well, or combination), distribution plumbing, coverage sprinklers or cannons, an industrial controller, a mesh network node, and an integration router. Where the water source requires a pump or pump starter, the controller provides a pump-control interface (a dry-contact start/stop signal, or a controller adapter card for larger starters) exactly as it interfaces any other controlled resource; a node on adequate pressurized supply requires no pump and populates no pump interface. Local systems operate independently when network coordination is unavailable and coordinate with neighbouring systems when the mesh network is operational. The intelligence in the system is resource management and coordination, not autonomous fire detection or targeting.

Layered Architecture

The system is organized as independent layers. Each layer is defined by the service it provides to the layer above and the interface it requires from the layer below. No layer depends on the internal implementation of any other. Technology at any layer may be substituted without modification to the layers above or below it, provided the interface contract is met. This is the architectural mechanism by which the specification absorbs new technology without revision to its core, and by which installations built decades apart remain interoperable.

The layering is organized along the lines of the OSI reference model, and the layer names below are given with their OSI positions. This is a **conceptual frame rather than a conformance claim**: the model is used because it is the one most readers already have, so that a network engineer, a controls integrator, and a fire officer can point at the same part of the system and mean the same thing. Nothing in this specification asserts conformance to OSI, and the architectural properties that matter — substitutability below the trust boundary, and dependencies that point in one direction only — do not depend on the layer numbering being canonical.

Physical and link layer (OSI 1–2). The radio, wire, or backhaul that moves bits between nodes. LoRa mesh via Meshtastic protocol and tactical-MANET-class StreamCaster MANET are the current reference classes. Ethernet, commercial LEO satellite service (Starlink class), government satellite transport (Starshield class or successor programs), cellular transport, and future radio classes are equally valid implementations of this layer. The specification imposes no requirement here beyond the ability to deliver coordination-domain traffic between participating nodes. Nothing above this layer knows or cares which technology carries the traffic.

Network layer (OSI 3). Participating nodes must share a common addressable coordination domain. A neighbourhood LoRa mesh satisfies this requirement. So does a set of remote installations joined by VPN over satellite backhaul. So does a hybrid in which residential mesh nodes and a federal facility on agency radio infrastructure are bridged through a gateway router. The requirement is the domain, not the medium. Routing, gateway selection, and addressing within the domain use standard mechanisms; this specification defines no novel protocols at this layer.

Envelope layer (OSI 4). The compact binary envelope and its delivery semantics. Delivery is best-effort by design; the system maintains no sessions and no connections. A missed message is a normal event rather than an error condition, and supply-pressure physics is the fallback coordination mechanism when messages do not arrive.

Vocabulary and trust layer (OSI 5–6). The closed, enumerated message vocabulary; quantized field encoding; cryptographic signing; and publication policy enforcement. This layer decides what leaves the node and attests to what does, and it is where the enumerated-message discipline becomes structural: a message whose verb is not in the enumeration has no representation and cannot be constructed. Data that the publication policy does not release cannot be transported, regardless of the capability of the underlying network.

Application layer (OSI 7). The coordination protocol — schedule negotiation, node state reporting, heartbeats, override handling, degradation behaviour — together with the event interface, incident command integration, the neighbourhood ledger, and after-event reporting. The coordination protocol is the invention. It is transport-independent by construction and shall remain so: no revision to this specification may bind it to a specific physical, link, or network technology. Consumers at this layer see coordination-domain participants and enumerated message categories. They do not see radios, transports, or local installation internals.

Two properties follow from this arrangement, and both are load-bearing.

Substitution. Everything at OSI 1–3 may be replaced — a different radio, a different mesh, a wire, a technology that does not yet exist — without modification above it. The trust layer and the coordination protocol cannot name a transport, and therefore cannot depend on one.

The publication chokepoint. Every outbound message, whatever application-layer function produced it, traverses the vocabulary and trust layer on its way out. Signing, scope resolution, and any rendering into an external exchange language occur there and nowhere else. The boundary between local data and published data is therefore enforced before any radio or wire is reached, and is independent of how the installation is networked. There is no path from the application layer to a transport that bypasses it; an implementation that provides one has not implemented this specification.

Control Architecture

The controller is industrial-grade PLC-based hardware mounted in a NEMA-rated enclosure with DIN rail mounting and standard terminal block wiring. This equipment class is chosen for reliability, long service life, availability through standard industrial supply channels, and compatibility with the installer pool of industrial electricians and controls technicians. Consumer-grade proprietary controllers are expressly outside the specification.

Controller requirements:

- IEC 61131-3 compatible programming environment or equivalent open programming interface
- Minimum sixteen digital I/O for pump control, valve control, flow sensing, and status indication
- Analog inputs sufficient for the installation’s sensing set, which may include supply pressure, expansion-tank or reservoir level, temperature, and optional acoustic or auxiliary sensing; the number of channels follows from the sensors actually fitted rather than a fixed minimum
- Serial or Ethernet interface for mesh node and router integration
- DIN rail mounting compatible with 35mm standard rail
- Operating temperature range appropriate for local installation environment

Example products meeting these requirements include the M-Duino series from Industrial Shields, the Kincony line, the Norvi controllers, and OpenPLC-compatible ESP32-based industrial controllers. Substitution of any controller meeting the capability requirements is compliant.

Enclosure

The controller, router, and associated interface hardware install in a NEMA 4X rated enclosure appropriate for outdoor and wet mounting. Standard hinged industrial enclosures meeting NEMA 4X (polycarbonate or stainless) satisfy the requirement. Enclosure size is not fixed: 16 by 20 by 8 inches is the reference size for a residential node, and any size that houses the internal panel layout and preserves the wet/dry separation is compliant. Detailed enclosure, mounting, grounding, labeling, and internal layout practice is specified in Annex A.

Radio Network Layer

The specification supports multiple radio infrastructure classes appropriate to different deployment environments. The specification does not prescribe a single radio technology because different participants have different operational requirements, different existing infrastructure investments, and different environmental conditions. Compliant installations select the radio class appropriate to their deployment context and integrate through the router layer.

LoRa mesh via Meshtastic protocol. Low-power mesh radio operating on the LoRa physical layer in the unlicensed ISM band allocated for the purpose in the jurisdiction of installation, using the Meshtastic protocol. The reference implementation operates in the 902–928 MHz band available in North America; installations elsewhere use the corresponding regional allocation (for example the 863–870 MHz band in Europe) at the power limit their regulator permits. Band and power are properties of the jurisdiction, not of the specification. **Operation is confined to unlicensed allocations.** A compliant node shall not operate on amateur radio spectrum. Amateur service rules require a licensed control operator and periodic station identification, restrict unattended automatic operation, bar communications that could reasonably be furnished through another radio service on a regular basis, and prohibit use where the operator holds a pecuniary interest — none of which an unattended, continuously operating, community-deployed node satisfies. Transmit power is bounded by the certification of the radio module as installed, not by the theoretical maximum of the allocation. Appropriate for residential and small commercial installations at neighbourhood scale. Low cost per node. Adequate coverage for typical suburban geometry. Suitable for homeowner-buildable installations following the RepRap tradition of specification-driven construction from documented parts.

Requirements:

- LoRa transceiver operating in the unlicensed ISM allocation of the installation jurisdiction at the maximum power that allocation permits; 902–928 MHz at 100 mW is the North American reference
- Meshtastic protocol firmware or protocol-compatible equivalent
- External antenna capability with weatherproof antenna mounting
- Battery backup sufficient for minimum 72 hours autonomous operation
- Serial or I2C interface to the local controller and router

Example products include the LILYGO T-Beam, the Heltec Wireless Stick Lite series, the RAK4631, and various other ESP32-based LoRa development boards packaged for industrial deployment.

Tactical MANET radio class. Agency-furnished mobile ad-hoc mesh data radios of the hardened interagency class (Silvus StreamCaster, Persistent Systems Wave Relay, TrellisWare TSM, and peers), deployed across federal land management, wildland firefighting operations, emergency medical services, law enforcement, and public safety broadly. Appropriate for federal facilities, remote installations on federal land, forestry cabins, ranger stations, and installations requiring range and reliability beyond LoRa mesh capability. Substantially higher throughput than LoRa enables coordination of larger installations with more comprehensive sensor and status telemetry. Compatible with existing agency radio infrastructure. The class is defined by capability — self-forming mesh, IP presentation, MIL-STD-810-class environmental qualification — never by vendor; any radio the furnishing agency fields and trusts is compliant at this layer.

Requirements:

- Tactical MANET-class radio (Silvus StreamCaster, Wave Relay, TSM, or interface-compatible peer)
- Self-forming mesh waveform with standard agency-compatible antenna configuration
- Integration through the router layer with layer abstraction from the coordination protocol

Interagency governance, adopted by reference: radio equipment operating on interagency wildland fire networks is governed by the National Interagency Incident Communications Division (NIICD) approved-equipment listings, and interoperability channel use by the National Interoperability Field Operations Guide (NIFOG). Nothing in this specification transmits on, contends for, or requires those networks; where agency-furnished radios are themselves NIICD-listed or incident-assigned, they operate under the furnishing agency's authority and the incident communications plan. This specification's coordination traffic rides its own unlicensed or agency-furnished data layers and never the fire voice nets.

Router Integration Layer

The router abstracts radio-layer differences from the coordination protocol. Participating installations exchange coordination messages through the router regardless of the underlying radio infrastructure. Homeowner installations with LoRa mesh nodes and federal installations with tactical MANET-class radios exchange the same coordination messages through the router layer. The router handles protocol translation, priority queuing, and network abstraction.

The router also serves as the policy enforcement point for publication decisions. Each installation configures its router to identify which log categories publish to the neighbourhood ledger and which stay local. This enables federal facilities, commercial installations, and privacy-conscious residential participants to participate in coordinated defense while maintaining appropriate boundaries on their internal operational data. Publication policy is specified in the Cryptographic Event Log section.

Router requirements:

- Radio layer abstraction supporting LoRa mesh and tactical MANET class at minimum
- Extensibility for additional radio classes as they are added to the specification
- Configurable publication policy engine with per-category filtering
- Cryptographic signing capability for outbound published messages
- Standard Ethernet or serial interface to the local controller
- Standard interfaces to supported radio hardware

The router is specified as functional capability. Reference implementations may run on industrial computing hardware, dedicated network appliances, or integrated within the controller platform. Substitution of any router implementation meeting the capability requirements is compliant.

Backhaul and Gateway Nodes

3.1 Satellite and External Backhaul

Installations and coordination domains may connect to external networks via commercial LEO satellite service (Starlink class, including compact low-power terminals suitable for expeditionary power budgets), government satellite transport (Starshield class or successor programs), or any transport presenting a standard IP interface to the network layer. The specification is transport-generic; named services are examples of the class.

Backhaul is an enhancement, never a dependency. A node or domain with no external connectivity coordinates locally, sequences, and performs its fire defense function in full. The backhaul extends the coordination domain and publishes the ledger; it is not required for fire defense. No revision may alter this.

3.2 Gateway Nodes

A gateway node is an ordinary node whose router additionally bridges the coordination domain to a backhaul transport. Gateway is a role, not a model; any node may serve as one. One gateway serves an entire coordination domain: because the coordination protocol's message categories are compact by design, an entire valley of participating nodes reaches the outside world through mesh hops to a single terminal. Multiple gateways are permitted for redundancy. Zero gateways is fully operational.

Gateway nodes advertise external reachability into the coordination domain using standard routing protocol mechanisms; nodes select among advertised gateways per normal route metric behavior, and failed gateways age out and traffic reconverges per those same mechanisms. **This specification defines no novel gateway election protocol.** At the IP tier this is standard router configuration; radio classes with native gateway mechanisms meet the requirement through the router integration layer.

The gateway is also a policy point: it sits at the trust layer boundary where domain traffic meets the outside world, and publication policy applies to it as to any router.

Water Supply and Reservoir

The water supply may be a dedicated cistern, connection to municipal supply, connection to a well system with adequate delivery capacity, or a combination. Minimum reservoir capacity for cistern-based installations is 2500 gallons, sized to sustain individual system operation during coordination scheduling for the duration of the anticipated ember-cast window. Neighbourhoods with reliable municipal supply may use municipal connection as primary source with a smaller reservoir for buffering. Remote installations without municipal supply require cistern capacity appropriate to their coordination participation and response scenario.

Water supply requirements:

- Delivery capacity is sized to the installation's coverage set and supply class. A node on pressurized supply (municipal or gravity) operates at the available service pressure and flow, from a single hose-bib sprinkler upward. A node on a pumped source (cistern or draft) is sized by its own pump and line selection; the pump and plumbing dictate valve and outlet sizing, not the reverse.
- Standard NPT threaded connections appropriate for the source and distribution outlets
- Backflow prevention consistent with local plumbing code
- Isolation valves for maintenance and emergency shutoff

Pump

The pump provides the flow and pressure required to operate the coverage sprinklers or cannons during activation. Gas or diesel drive is preferred to eliminate dependence on grid electric power during events. Electric pumps with adequate battery backup or generator support are permitted where fuel-driven pumps are impractical.

Pump requirements:

- Self-priming, capable of lift from reservoir mounting configuration
- Rated flow minimum 40 gallons per minute at 80 psi discharge pressure
- Fuel drive: gasoline or diesel, minimum four hour continuous runtime on onboard fuel
- Electric drive: 240V single phase or 208V three phase, appropriate motor protection
- Remote start capability compatible with PLC control output

Example products include the Honda WX10 and WX15 series, the Predator 6.5 HP gas water pumps, various trash pump configurations rated for continuous operation, and industrial centrifugal pumps from Grundfos, Goulds, or equivalent manufacturers for electric installations.

Coverage Distribution

Coverage distribution places water on structure surfaces and adjacent receiving sites where embers land. Standard configurations include roof-line sprinklers, gutter-line spray heads, perimeter rotating sprinklers, and directable cannons for larger structures or specific coverage requirements. The specific configuration depends on structure geometry, roof material, adjacent vegetation, and property topology.

Coverage distribution requirements:

- Roof-line coverage sufficient to maintain wet surface across full roof area
- Gutter coverage sufficient to prevent ember accumulation in gutter debris
- Perimeter coverage extending minimum ten feet from structure walls
- Zone valving allowing selective activation of coverage areas
- Materials rated for radiant heat exposure during fire events

Coordination Protocol

Local Autonomous Operation

Each local system operates autonomously when the mesh network is unavailable or when coordination messages are not received within specified timeout intervals. Autonomous operation defaults to conservative water use appropriate for extended individual protection without regard to neighbourhood coordination. The system does not fail when isolated; it operates at individual capability without the coordination advantages the mesh network provides.

Coordinated Operation

When the mesh network is operational and coordination messages are being exchanged, participating systems synchronize their activation schedules. The coordination algorithm allocates operating time across participating systems such that aggregate water draw stays within the sustainable supply capacity while ensuring that no participating structure has an extended gap in coverage during the ember-cast window.

The reference coordination algorithm operates on the following principles. Each participating node reports its water source category, current supply state, coverage pattern, and current activation state to the mesh network through its router. The distributed algorithm computes a rotating schedule that keeps sufficient nodes active to maintain neighbourhood coverage while allowing individual nodes to conserve water and pump runtime during their off-cycles. Schedule recalculation happens continuously as node states change.

Graceful Degradation

The specification anticipates realistic degradation patterns. Individual nodes may fail during operation. Water supplies may be interrupted. Mesh network connectivity may vary. Fire department override may or may not be available. The system continues providing value at each degradation level. A node with failed water supply but functional mesh capability continues serving as a network repeater. A neighbourhood with reduced participation continues providing coordination for the participating subset. A neighbourhood operating without professional response integration continues operating on its autonomous coordination schedule.

Event Interface and Message Category Enumeration

2.1 The Event Interface

The system exposes one outbound integration port: the event interface. It is a read-only tap of the system’s internal event messaging — the same enumerated messages the system already generates for its own operation. Through this interface, external systems may be informed that an activation has started, that a zone is cycling, that supply pressure is degraded, that an event has ended, that the system requires maintenance.

The event interface is deliberately narrow:

- **Outbound only.** External systems listen. The event interface accepts no inbound commands and transports no inbound payloads. Control of the installation remains with the installation’s own controller, its operator, and — where integrated — the incident command interface defined in the main specification.
- **Enumerated messages only.** The interface carries the message categories enumerated in this specification and nothing else. It is not a general-purpose bus.
- **Lightweight by construction.** Specified as a simple application-layer event protocol suitable for serial, Ethernet, or dry-contact presentation. A home automation system, generator controller, building management system, or hand-built display panel may consume it with minimal integration effort.

A smart home may listen to the Firefighter. The Firefighter does not join the smart home.

2.2 Enumerated Message Categories

Published message categories are enumerated by this specification. The coordination mesh transports enumerated categories only. A message without a category defined in this section has no representation in the coordination protocol and cannot be transported by a compliant installation, regardless of the publication policy configured at the router.

Residential and light commercial tier:

- System status: heartbeat, readiness, fault, maintenance-required
- Distress: node identity, position, timestamp — published on operator activation of the hardware emergency control; deliberately minimal
- Node position: GNSS-derived location; coarse-position-only publication is a supported policy option
- Activation events: event start, event end, trigger source class
- Zone state: zone open/close transitions, cycle schedule position
- Supply state: pressure, flow aggregate, reservoir level where equipped
- Coordination traffic: schedule negotiation, node state reports, override acknowledgments
- Command broadcast: directive class (cease water operations, resume, priority protect, broadcast alert), scope filter, issuing authority, expiry — see Section 2.4
- Energy events: power source transitions, battery state, low-voltage disconnect
- Environmental telemetry: temperature, wind speed and direction, humidity, ember detection *events* where equipped

Expeditionary tier — additional categories:

- Pump state: running/stopped, RPM class, load state, fuel level or acoustic burn-rate forecast, start/stop command acknowledgment where pump control is supported
- Distribution state: lay pressure, flow, burst/loss detection events

Government and high-specification commercial tier — additional categories:

- Thermal detection events: detection occurrence, bearing class, confidence class — detection *events*, not imagery
- Triage state: internal triage strategy selection, contingency template activation
- Extended sensor telemetry as defined in the applicable tier annex

Expressly non-transportable at all tiers:

- Video and imagery of any kind
- Audio of any kind, including raw or compressed audio from acoustic machinery sensors; acoustic sensing publishes derived machinery state events only
- Physical security data: camera-derived data, motion detection intended for security purposes, access logs
- Personnel data of any kind

These categories are not merely “typically not published” — they have no message category and no representation in the coordination protocol. There is no compliant configuration in which they transit the mesh.

2.3 Local Sovereignty

Local input/output belongs to the operator. Nothing in this specification prevents an operator from connecting cameras, motion detectors, microphones, or any other equipment to spare controller inputs to trigger local behavior — starting a local recording, sounding a local alert, opening a local valve. Such integrations are compliant. They terminate at the router. The router’s publication policy engine has no category in which to place them, and the coordination protocol has no message type to carry them.

The rationale is stated plainly so future revisions preserve it. First, bandwidth: the residential radio class cannot carry rich media, and the specification treats that constraint as a design asset rather than a limitation to be engineered away — the same message discipline that enforces the mission boundary is what allows an entire coordination domain to share a single thin backhaul. Second, attack surface: data that is never transported cannot be intercepted, and a sensor that is not networked cannot be reached; the residential installation’s security posture rests on carrying almost nothing. Third, mission: this is a fire defense specification. Capacity for general-purpose home monitoring is architecturally adjacent and deliberately unbuilt. Operators with sophisticated custom systems should find in this specification a model to learn from, not a platform attempting to absorb them.

2.4 Command Broadcast

Incident command may issue domain-wide directives as connectionless broadcasts — fire-and-forget datagrams to the coordination domain, with no acknowledgment round and no per-node addressing. The motivating case: an engine company drafting from a municipal main needs the main clear immediately; the automatic pressure-sag response yields the same result but across sensing and convergence lag. The broadcast is the human jumping ahead of the control loop.

Broadcast semantics:

- **Delivery is best-effort by design.** Nodes that receive the directive comply; nodes that miss it continue local operation and yield through the normal pressure-sag response — the physics is the fallback delivery mechanism. Graceful degradation requires no additional machinery.
- **Scope filters ride in the message:** by supply class (the primary case — municipal-supplied nodes cease while cistern, pond, and reservoir nodes continue protecting), by zone, or by geography. Nodes evaluate the filter locally against their own configuration; no central registry exists. Nodes outside the filter, or without the resources the directive concerns, ignore it.
- **A cease-operations directive to a node with stored water is a source switch, not a shutdown:** the node drops external draw and continues coverage on its own reserves. The directive constrains draw, not defense.
- **Directives carry expiry, not release.** Every command broadcast has a time-to-live and must be refreshed by the issuing authority; on expiry, nodes resume automatically. A stale silencing order cannot outlive the attention of the commander who issued it.
- **Directives are signed and audited:** issued under the incident command interface's existing override authority, verified at the trust layer, and logged to the ledger by every complying node. The after-event record shows what was ordered, by whom, when, and what each node did.
- Operator hardware controls are unaffected, per the main specification's override provisions and Charter Clause 7.

2.5 Single Command Surface

All enclosure classes present through a single command surface. A residential node on a hose bib and a tactical node at a pond appear in the same operational picture, distinguished by capability metadata carried in their status messages, never by separate applications. No enclosure class shall require class-specific command software. The incident command interface, including its presentation into existing agency systems (TAK-class and ICS software per the main specification), sees a coordination domain — one protocol, one message enumeration, one pane of glass. Class differences end at the enclosure wall.

2.6 Common Exchange Languages (Adopted at the Boundary)

The tactical, incident-command, and public-alerting worlds already maintain common exchange languages, and this specification adopts them at its publication boundary rather than inventing competitors:

- **Cursor-on-Target (CoT) / TAK protocol** — the common operating picture grammar of the TAK ecosystem and the tactical MANET world. Node presence, status, position, detections, and events publish outbound as CoT events with appropriate type codes; this is the formal identity of the "TAK-class publication" named elsewhere in this specification.
- **Common Alerting Protocol (CAP)** — the OASIS/IPAWS standard for public alerting. Where a coordination domain's detection layer contributes to community warning, the domain publishes outbound as CAP.
- **SIA Contact ID / DC-09** — the fire-alarm industry's device-status and trouble grammar, for consumers that are alarm-monitoring systems rather than tactical or public-alert pictures.
- **TIA-102 (P25) data services** — the exchange grammar of public safety trunked data, noted per the P25 section; integration, where an agency elects it, presents through the same translation boundary.

Translation is outbound-only and occurs at the router. The governing principle is adoption by reference: wherever a destination system already has an established standard for a concept the domain must express, OSF publishes in that standard rather than inventing a synonym — CoT toward a tactical picture, CAP toward public alerting, Contact ID toward alarm monitoring, and additional destination standards as they arise. The publication policy layer renders the closed OSF vocabulary into whichever exchange language the consumer speaks, at the gateway boundary. Inbound command remains exclusively the closed, signed, enumerated OSF command set — external systems read the domain in their native grammar but command it only through ours. The security

architecture is preserved by construction: the mesh never carries a rich external format, and no external language can express a command the vocabulary does not contain. The mapping tables (OSF↔CoT, OSF↔CAP, OSF↔Contact ID, and further destinations as adopted) are an specification appendix deliverable, maintained under charter governance alongside the vocabulary itself.

Incident Command Interface

The specification requires incident command interface capability sufficient for professional response integration during multi-jurisdiction wildland-urban interface events. The interface presents unified operational picture across all participating systems regardless of underlying radio infrastructure. Incident commanders arriving on scene operate one interface to see and coordinate with the entire distributed defense operation.

Interface Presentation

The interface presents information consistent with Incident Command System conventions. Standard incident command information appears in standard positions. Water system status integrates into overall tactical picture. Override authorities are available with appropriate confirmation for consequential actions. Radio-class differences abstract away so the commander sees participating nodes as a unified list with capability metadata rather than as separate networks requiring separate management.

Override Authorities

The interface supports commander override of automated coordination. Priority activation of specific structures. Exclusion of zones where fire department personnel are operating. Modification of aggregate draw when municipal supply is stressed. Broadcast alerts to all participating systems. Suspension of automated operation during commanded operations. Command authority mapping respects actual authority structures at events. Homeowners retain override of their own systems even during commanded operations, but coordination happens through the incident command structure when it is operating.

Existing System Integration

The interface supports integration with existing incident command systems where they are deployed. Federal wildland incidents use specific ICS software and coordination systems. Municipal departments use various commercial and open systems. The interface can present its coordination picture into these existing systems so the water defense operation appears as one more resource type in the commander’s overall tactical picture rather than as a separate system requiring separate attention.

Optional Deployment

Neighbourhoods may deploy the specification without incident command integration. Fire departments may choose not to integrate with participating neighbourhoods. The system operates at its designed capability with or without integration; integration is an enhancement rather than a requirement. Where deployed, the integration operates through the router layer without requiring modification of existing agency infrastructure.

Cryptographic Event Log

The specification requires cryptographically verifiable operational logging sufficient for post-event trust establishment. The log architecture uses change log format with periodic snapshots, balancing bandwidth constraints during active events against public-facing verifiability after events.

Local Operational Log

Each participating system maintains its own operational log at whatever resolution its local storage supports. The log captures everything the system knows about its own operation: trigger events, sensor readings, PLC state changes, coordination messages sent and received, override commands accepted, water flow measurements, pump runtime, PLC errors, and maintenance events. High-resolution local record lives on the local system.

Snapshot Publication

Periodic snapshots roll up the local log into committed states published to the neighbourhood ledger. Snapshot frequency depends on operational context. During normal operation, daily or weekly snapshots capture routine

status. During active events, snapshot frequency increases to whatever the mesh bandwidth supports, capturing operational state changes as they occur. Snapshots are compact by design: minimal traffic when nothing is changing, summary state when changes are occurring.

Log Data Categories

The specification defines log data categories sufficient for unambiguous filtering by publication policy. Each installation configures which categories publish to the neighbourhood ledger and which stay local.

- Fire defense operational data: system activation, coordination participation, water use, coverage status. Minimum publication requirement for meaningful participation.
- Environmental sensor data: temperature, wind, humidity, ember detection when equipped.
- Coordination messages: mesh network exchanges, override commands, scheduling decisions.
- Physical security data: cameras, motion sensors, access logs. Typically not published.
- Personnel data: any information about individuals present at the installation. Typically not published.
- Installation-specific operational data: proprietary process data, agency internal operations. Typically not published.

The message category enumeration in the Event Interface section governs what the coordination mesh can transport; the categories above govern what the ledger records and what publication policy filters. Categories marked expressly non-transportable in the enumeration have no representation on the mesh regardless of ledger configuration.

Publication Policy

Each installation's router enforces publication policy. Federal facility routers filter camera feeds, personnel data, and installation-specific operational data. They publish the fire defense subset. Commercial facility routers filter based on what they need to protect. Residential routers publish according to homeowner preferences. Publication policy is configurable at installation and modifiable by the installation operator without requiring changes to the coordination protocol.

The specification defines minimum publication requirements for meaningful participation: activation events, water use aggregate, coordination participation acknowledgment, and system status heartbeat. Everything else is at the participant's discretion. This ensures the ledger has enough data to be useful for post-event reconstruction while respecting the flexibility that different participants need.

Cryptographic Verification

Snapshots are cryptographically signed at creation and appended to a chain that makes tampering detectable. Each entry references the previous entry's hash. Signing keys are held by participating systems and by trusted witnesses appropriate to the neighbourhood's trust structure. Fire departments, insurance carriers, and community coordinators may participate as witnesses. The ledger is distributed across nodes so no single point of control can rewrite history.

The specification identifies capability requirements for the ledger technology rather than prescribing specific implementation. Cryptographic signature verification, append-only chain structure, distributed storage, and threshold witnessing are the required capabilities. Implementations may use established distributed-ledger technologies or purpose-built ledger implementations meeting the capability requirements.

Public Access

The neighbourhood ledger provides queryable access to published snapshots. Time range selection, node selection, event category filtering, and appropriate detail level are supported through the query interface. Consumers of the interface build their own report views appropriate to their needs. Insurance systems, incident command systems, community coordination interfaces, and specification refinement research all access the same underlying ledger data through views appropriate to their purposes.

After-event reports generate from ledger queries rather than from operator narratives. The reports show what happened when, in what sequence, with what system states, and what outcomes resulted. Because the ledger is verifiable, the after-event reports carry the trust weight of the cryptographic verification rather than depending on the operators' credibility.

4.1 Deployment Class

The expeditionary configuration serves forestry, National Guard, and emergency management deployments to areas without installed coverage. Current structure-protection practice deploys pump-and-hose kits — a portable pump drafting from a river or pond, hose lays, tripod sprinklers, a fuel supply — running until dry or until a crew returns to check. The expeditionary node ties into this practice without modifying it: dropped alongside the kit, it provides real-time pump-site intelligence, deliberate water and fuel scheduling, and coordination-domain participation, with zero contact with agency pump equipment required.

4.2 Capabilities

- **Pump-site telemetry:** discharge pressure and flow via an inline sensor spool clamped into the hose lay; engine state via non-contact acoustic sensing (derived events only, per Section 2.2); fuel forecasting from burn-rate inference. Hose bursts, pump stoppages, and fuel exhaustion become immediate events instead of discoveries on the next crew visit.
- **Duty-cycled protection:** zone valves in the distribution lays allow the sequencer to time-share pump runtime across coverage points, extending a fixed fuel load across the ember-cast window. Continuous run-to-dry operation is replaced by deliberate consumption.
- **Pump control interface:** published in advance of industry support — 12V nominal control convention compatible with existing portable-pump control panel practice, start/stop/state message categories, relay-drive electrical interface. Current-generation portable pumps are sense-only; the interface definition provides procurement language through which agencies can specify coordination-interface pump control in future tenders. Battery-electric portable pumps, when available, integrate under the same interface; duty-cycled sprinkler protection is the operating mode that makes battery-electric portable pumping viable at current energy density.

4.3 Automated Fire Lookout Configuration

The composite kit — expeditionary node, pump and hose kit, weather station, and (at government tier) thermal detection payload — constitutes an automated fire lookout that defends its own site. The historical staffed lookout could see fire but not survive it; this configuration wets its own perimeter. Local weather measurement feeds wet-cycle scheduling directly and provides incident command with spot weather at the point of greatest forecast uncertainty. Detection events ride the coordination protocol per Section 2.2; imagery, where an agency requires it, rides agency radio infrastructure as agency traffic outside this specification’s enumeration. Agency security or surveillance assets deployed adjacent to a node coexist on the same network without being absorbed by this specification.

4.4 The Coordinated Defense Module

The expeditionary configuration composes into a deployable resource type assembled entirely from existing fire service elements: one Type 6 engine, one trailer of cased units, a crew of three to four, and a UAS operator. Nothing in the module is new — the crews, engines, typing conventions, and dispatch mechanisms already exist; the module is a new line item in the resource catalog, orderable and demobilized like any typed asset.

Employment concept — seed and move. The module arrives ahead of the fire at a threatened structure cluster and, through the open-lid commissioning ritual, stands up persistent nodes where structure protection would otherwise stand people. Because the nodes hold ground autonomously, the module does not stay: it leapfrogs to the next cluster and seeds it, laying persistent protection at multiple communities per operational period. The staffing arithmetic inverts from crews-per-site to sites-per-crew. At demobilization, cases are inventoried, signed, and recovered per the joint-inventory doctrine — the resource that lays resources, and takes them home.

Network self-assembly. Every unit carries the common mesh radio regardless of radio tier. Where participating residential systems exist, module nodes join the standing coordination domain on power-up — no integration step, by construction. The exchange runs both directions: the community’s mesh delivers established topology, supply intelligence, and a coverage map to the arriving module; the module delivers command presence, detection, backhaul, and tactical nodes placed at exactly the gaps the domain census shows. **Prepared communities are cheaper to defend, and the network is how the module knows it on arrival** — residential adoption converts measurably into tactical capacity.

Range convergence. Coordination density follows interface density by design: the system’s hydraulic reach (hose-

lay scale), realistic terrain mesh range, and the structure-spacing envelope at which coordinated defense matters all converge at neighborhood scale. Field heuristic: *if a hose could reach it, the mesh reaches it.* Sparse and remote deployments operate as autonomous clusters with reporting reach, per the graceful-degradation architecture; the sweet spot is the interface, which is where the problem lives.

4.5 Prepositioning

Expeditionary units are designed for palletized warehouse storage and drop deployment. **Prepositioned units shall require no external activation, registration, provisioning service, or connectivity to enter service.** A unit sealed in its case enters service years later by being opened, connected, and powered. Warehouse maintenance is limited to battery inspection and commodity-part replacement on standard storage intervals. A gateway kit — satellite terminal, power supply, solar panel — is a single additional pallet line item per coordination domain.

4.6 Terrain-Anchored Deployment and Non-Destructive Suppression

The preceding sections defend structures. This section defines the system's employment against fire at terrain scale: holding ground rather than protecting buildings. It is the highest tier of expeditionary employment and carries the strictest command and bounding conditions.

Terrain-anchored placement. A node's target may be the land, not a structure. Fuel corridors, drainages, chokepoints, saddles, fuel breaks, and field edges — the ground a fire must cross — are legitimate node placements, wetted as coordinated defensive elements ahead of a front. A structure at the mouth of a drainage that cannot be defended at its own walls (for example, one taking direct radiant load from a crowning slope at close range) may nonetheless be defended by denying the fire the corridor upstream.

Pre-surveyed siting. Deployment locations are sited in advance, not improvised under fire. A century of fire history, fuel and terrain data, prevailing wind, and satellite imagery identify where a given landscape burns before any specific fire exists. Chokepoints, wet-line corridors, and green-island/landing-zone sites are surveyed, walked, and mapped in the off-season, embedded in the Community Wildfire Protection Plan, and cached against the named corridors they will hold. Model-and-imagery-based siting is decision support; a qualified person commands placement. The fire executes a plan drawn when the ground was quiet.

Green islands. A pre-wetted patch of ground ahead of a front serves two functions at once: a break point where the fire arrives with reduced energy and can be stopped or steered, and — once committed and safe — a helicopter landing zone. The system generates tactical terrain and aviation infrastructure from the same water it uses to defend.

Wet line as reversible fireline. A continuous wetted strip is a constructed fireline that builds itself faster than hand or dozer line and is held by the nodes rather than by crews. Its tactical function is identical to fuel-removal line — deny the fire the next foot of receiving fuel — but achieved by temporarily raising fuel moisture rather than permanently removing fuel. The line exists when needed and evaporates when the threat passes; the vegetation lives, the ground is unscarred, and the community is not destroyed to be saved. This reversibility is the doctrine's central advantage: the political and ecological objections that limit destructive line construction (bulldozed property, felled trees, speculative damage) do not apply. Wetting a fine (1-hour) fuel below ignition threshold is forestry-manual-level calculation; the supply required to hold a narrow line against a surface fire is small, and modest static sources (a drum uphill, a small cache, a tender-fed porta-tank layer) suffice.

Passive until triggered. Unlike an active hose lay, which commits a crew for its entire duration, a terrain-anchored network is emplaced and dormant — present at the dry corridor at negligible standing cost, doing nothing on the great majority of days, and deploying automatically alongside the static residential network only when conditions and a front arrive. This separation of *presence* from *staffing* is what makes landscape-scale coverage economically possible where continuous hose-kit staffing is not.

Bounding conditions (mandatory). - This is a surface- and moderate-intensity-fire tactic, and a flanking/backing-phase tactic. A wet line denies fire spreading by contact with surface fuel. It does **not** hold a wind-driven crown fire spreading by radiant heat and long-range spotting. Doctrine shall not claim otherwise, and no crew's egress shall be staked on a wet line as though it were a guaranteed barrier. - The wet line is best-effort infrastructure and may be lost like any line; it buys time and probability, not certainty. - Employment is crew-directed and human-commanded. The system is a tool a qualified incident commander or division supervisor wields; it does not conduct autonomous suppression. Where automated action and command judgment meet, command prevails (Charter Clause 7). Placing "containment" without a qualified human owning the tactic is expressly non-compliant.

The force it serves. This doctrine brings a technical advantage to the existing wildland firefighting community — it does not replace it. The line is held by nodes and patrolled, tended, and commanded by people: seasonal hand-crew labor matched to light distributed work (patrol, valve-tending, spot-chasing, node-minding), with qualified leaders on

Type 6 engines running the tactical picture. It is easier and lower-risk than cutting hand line and requires fewer specialized bodies, because the boxes hold the line the crews would otherwise dig and man. When the fire is gone, the crew leaves and the ground is intact. This is a force multiplier for the Pulaski community, built by one of its own — not autonomy displacing the firefighter, but reach and leverage handed to the firefighter who already does the work.

Installation and Maintenance

Installer Qualifications

Installation is within the capability of any competent industrial electrician, licensed plumber, or capable homeowner with appropriate mechanical and electrical skills. The specification does not require licensed installers beyond what local building codes require for the plumbing and electrical work involved. Retrofit of existing residential irrigation systems is within the capability of any licensed irrigation contractor. Federal facility installations follow the agency’s normal contracting and installation practices with the specification providing the technical requirements.

Building Code Compliance

Installations comply with local building codes for plumbing, electrical, and structural work. Backflow prevention meets local water district requirements. Electrical installation meets the applicable National Electrical Code requirements or local equivalent. Structural mounting of roof-line coverage meets local roofing and structural requirements. Standard permitting processes apply.

Operator Responsibility

The specification treats the installation as operator-owned equipment operating under operator responsibility. Maintenance, testing, component replacement, and continued operational readiness are operator obligations. This is the smoke alarm model of property protection: the equipment is installed, the operator maintains it, the system operates when needed if the maintenance has been kept up. No subscription service, no monitoring contract, and no ongoing external obligation is required for the equipment to function.

Cost Estimation

Total installed cost varies substantially with local labor rates, existing infrastructure, specific component selections, and radio class selected. Representative cost ranges for new construction from bare property, residential installation with LoRa mesh:

- Reservoir and plumbing: $1,500 to $3,500 depending on capacity and configuration
- Pump: $500 to $1,500 depending on drive type and flow capacity
- Controller, enclosure, router, and LoRa mesh node: $700 to $1,500
- Coverage distribution: $800 to $2,000 depending on structure size and complexity
- Installation labor: $1,000 to $3,000 depending on local rates
- Total residential range: $4,500 to $11,500 for complete new installation

Federal facility and remote installation costs vary substantially based on tactical radio infrastructure requirements, existing radio investment leverage, and installation environment. Facilities with existing tactical MANET infrastructure add the coordination capability at substantially reduced radio cost. Facilities without existing infrastructure incur higher radio costs offset by capability appropriate to the deployment scale.

Retrofit of existing residential irrigation systems reduces cost substantially. Existing controllers may accept mesh coordination modules without replacement. Existing water supply and distribution infrastructure may require only the addition of coverage-appropriate sprinkler heads and zone reconfiguration. Retrofit installations may be completed for $1,500 to $4,000 depending on existing infrastructure.

Licensing

This specification is released under permissive open licence. Any operator, contractor, agency, or manufacturer may implement compliant installations without licence fees or ongoing royalty obligations. Improvements to the specification through community development are welcomed and integrated into subsequent versions.

Governance Charter

5.0 Adoption by Reference (standing rule)

External standards, codes, and regulatory regimes named anywhere in this family — NWCG publications, NFPA codes, UL standards, FAA regulations, NIICD listings, NFDRS, Section 508, AWWA practice, and all others — are **adopted by reference to their current edition as amended by their issuing authority**, never copied out or recreated. Where any document in this family paraphrases or names a value from an external regime, the external regime's current text governs and this family's paraphrase is descriptive only. This is the same mechanism law uses to remain dynamic: if the issuing authority changes a threshold, an altitude, a listing, or a table, this specification follows automatically and requires no amendment. This family invents only in the space no authority has claimed; everywhere an authority exists, we defer to it, by reference, as it evolves.

5.1 Stewardship Model

The specification is maintained in the manner of open network protocol standards. Anyone may propose a revision. Proposals are evaluated in public against this charter. Adoption requires demonstrated need within the specification's mission; novelty alone is not need. The versioned, cryptographically chained change log defined in the main specification is the governance record: proposals, dispositions, and the reasoning for each are committed to it, so that the standard's history is verifiable by the same mechanism as its installations' operations.

Proposals are evaluated first by locating them in the layered architecture. A proposal located at the physical, link, or network layer is typically already accommodated by the existing interfaces or already excluded by layer discipline, and requires no revision to adopt or reject. Only proposals touching the vocabulary and trust layer, the coordination behaviour, or the message enumeration modify the invention, and these receive the fullest scrutiny.

5.2 Charter Clauses

All revisions are evaluated against the following clauses. A proposal in conflict with any clause is not adopted, however capable the technology it introduces.

Clause 1 — Buildability. The core system shall remain buildable from commodity components available through standard industrial supply channels by a competent individual, electrician, or controls technician. No revision may make the core system dependent on components available only from a single manufacturer or only through restricted channels.

Clause 2 — No mandatory dependency. No revision may introduce a mandatory dependency on external network connectivity, cloud services, subscription services, license servers, or any ongoing external obligation for the core system to perform its fire defense function. The system operates when isolated. This clause is absolute.

Clause 3 — Extension by module. New capability enters the specification as an optional module with its own conformance section. It does not enter as a modification of core requirements. The core remains small; the periphery may grow.

Clause 4 — Non-adoption is not prohibition. Rejection of a proposal bars no one from building it. Extensions not adopted into the standard are simply not part of the standard: they receive no message category, no conformance language, and no compatibility guarantee. Their builders remain free to build.

Clause 5 — Mission boundary. The specification defines a coordinated fire defense system. Revisions extending it toward general-purpose home automation, security monitoring, or surveillance are outside its mission and are not adopted, without regard to technical merit. The message category enumeration of Section 2 may be extended only for fire defense purposes.

Clause 6 — Layer discipline. Revisions respect the layered architecture. New physical, link, and network technologies are welcomed at their layer and shall not impose requirements on the layers above them. A new sensor generation that assumes internet connectivity does not thereby make the specification assume it; the specification's answer to new devices is its existing interfaces.

Clause 7 — Operators prevail. Automation in this specification manages resources; it does not overrule operators. Where automated policy and operator action conflict at the point of use, the operator prevails, and the system's role is to make consequences visible — through status indication, published energy and state events, and the operational log — not to refuse. This extends the main specification's existing principle that homeowners retain override of their own systems even during commanded operations. No revision may introduce an automated lockout of an operator's physically present resources or controls.

5.3 Worked Example

A manufacturer introduces a DIN rail device with integrated Bluetooth provisioning. Under this charter: the device is not barred from use — an installer may deploy it where it meets the capability requirements of its component class (Clause 4). Bluetooth does not become part of the specification, because no demonstrated need within the mission requires it (Clause 5) and the core gains no new dependency (Clause 2). If a future revision finds genuine need — say, commissioning tooling — it enters as an optional module with its own conformance section (Clause 3), and the core system remains fully buildable and operable without it (Clauses 1, 2).

5.4 Amendment of this Charter

The charter itself may be amended only by the same public proposal process, recorded in the change log, and no amendment may weaken Clause 2 or Clause 7.

Acknowledgment

The specification represents synthesis of existing engineering practice across residential sprinkler installation, industrial control systems, mesh network protocols, tactical mobile ad-hoc networking, cryptographic ledger technology, and volunteer and professional fire service operations. It does not depend on proprietary technology and does not require novel engineering development. It is arithmetic and integration applied to a defensive coordination problem that has not previously been addressed at the scale and across the participant range that actual wildland-urban interface fire events involve.

Appendix C — Wire Format and Vocabulary

Consolidated into this specification; previously issued separately as Annex C. This appendix is the authority for the closed message vocabulary, envelope layout, and the outbound cross-reference to external exchange standards.

Envelope

The wire format is a compact binary envelope sized to the most constrained transport in the family. LoRa mesh payloads live in the ~200-byte range; the envelope must fit it with room for a body.

Fixed header, every message: **version · message class · node ID · sequence number · timestamp · signature.**

The signature is the large field (Ed25519 class, 64 bytes) and is non-negotiable — the trust layer rides in the envelope. Realistic body budget after header and signature: **60–100 bytes.** That number is the discipline; every verb must fit it.

Shape Decisions (Fixed Now)

1. **Enums everywhere, strings nowhere.** Human-readable names live in the specification document, never on the wire. This serves bandwidth and, more importantly, the mission boundary: a protocol with no string type cannot smuggle rich payloads. The absence of a general-purpose payload field is a structural property, not a policy.
2. **Quantized classes over precise values.** Coordination needs “pressure sagging,” not 47.3 psi. Readings on the wire are single-byte quantized classes; full-resolution values stay in the local log per the main specification.
3. **Capability bits in the heartbeat.** Each node’s periodic heartbeat carries capability flags (supply class, stored-water presence, pump control, detection payload, gateway role). This is how the single command surface knows what each node is without asking, and how command-broadcast scope filters resolve locally without a registry.
4. **The vocabulary is closed and short.** Additions go through charter governance. Smallness is a protected property of the protocol, not a v1 limitation.

V1 Verb List

Grouped per the message category enumeration (Section 2.2):

Group	Verbs	Notes
Presence	HEARTBEAT	status flags, battery class, supply class, capability bits, coarse position
State	ARM, DISARM, ACTIVATE, DEACTIVATE, ZONE_OPEN, ZONE_CLOSE, FAULT, MAINT_REQ	FAULT carries a fault-code enum

Supply	PRESSURE, FLOW, LEVEL	quantized single-byte classes
Coordination	SCHED_STATE, SCHED_PROPOSE, YIELD, RESUME	schedule negotiation per main specification
Command	CEASE_OPS, RESUME_OPS, PRIORITY_PROTECT, BROADCAST_ALERT	scope filter, authority, expiry per Section 2.4; short, audited vocabulary
Safety	DISTRESS	node, position, timestamp — deliberately minimal
Events	DETECT, ENGINE_STATE, ENERGY	detection events (source/bearing/confidence class); acoustic-derived machinery state; power transitions
Ledger	SNAPSHOT_COMMIT	hash chain link; the payload is the commitment

Roughly two dozen verbs. The full annex will define field layouts per verb and four worked byte-level examples: HEARTBEAT, ZONE_OPEN, CEASE_OPS, DISTRESS.

Why This Is Most of the Software Design

The vocabulary list is the data dictionary. An implementer inherits a closed verb set, field definitions with quantization decided, an envelope with the byte budget pre-fought, and a charter that bounds what may be added. The enum table compiles nearly directly to a header file; the verb list is the dispatch switch; the state transitions described in the main specification are the state machine. Two independent implementers working from this table produce interoperable nodes without coordination — convergence by construction, which is the property that makes distributed community implementation work.

Queued for full drafting during the community development window, alongside reference implementations. Constraints above are fixed; encoding details are not.

Vocabulary Cross-Reference (Adopted Terms)

The OSF vocabulary is small and closed. Where a concept it must express already has an established term in the two domains this system publishes toward — the tactical world (Cursor-on-Target / TAK), the fire-alarm world (SIA Contact ID / DC-05, transported by DC-09), and the public-alerting world (Common Alerting Protocol / CAP, for community warning) — OSF adopts the established term rather than inventing a synonym. The principle is general: where a destination system has an established standard for a concept, OSF publishes in it; CoT, Contact ID, and CAP are the destinations developed first, and others are added the same way as they arise. **Rule of precedence: where the same concept exists in all three (OSF, CoT, Contact ID), the tactical (CoT) form is the OSF standard**, because CoT is the grammar OSF publishes into at the incident-command boundary and the one a coordinating system will most often translate against. Contact ID governs where the concept is native to alarm practice and absent or weaker in CoT (device trouble states). **No OSF output is ever rendered as an alarm-class Contact ID event.** Contact ID rendering is limited to supervisory, trouble, and status codes — the device reporting its own condition to an alarm-monitoring consumer. The system does not signal a fire condition to a monitoring station, and does not initiate a response; that function belongs to a listed fire alarm system and is deliberately outside this specification. This adoption applies to **outbound status and telemetry only**; the inbound command enumeration remains closed, signed, and OSF-native — no external vocabulary is adopted on the command path, by design (§ security model).

A structural finding falls out of the mapping: **almost every cross-domain match lands on the outbound/telemetry vocabulary, and almost none on the command verbs.** Status concepts — heartbeat, position, fault, low battery, tamper, detection, time/expiry — are universal and already standardized elsewhere; the command verbs (WET, terrain-line actions) are domain-unique and have no external equivalent to adopt. This is the security model expressing itself in the vocabulary: the words the system *speaks outward* are shared; the words it *accepts inward* are its own.

Outbound status / telemetry — adopted forms:

OSF concept	CoT (tactical)	Contact ID (alarm)	OSF standard adopts

Node presence / heartbeat	`<event>` with `stale` soft-state; periodic self-report	Periodic test (E602)	**CoT** — stale/expiry soft-state model (triplicate → tactical)
Node identity	`uid` (persistent)	Account + zone number	**CoT** — persistent `uid` (triplicate → tactical)
Position / location	`<point lat lon hae ce le>`	(none; panel is fixed)	**CoT** — native geospatial
Time / event validity	`time` / `start` / `stale`	Message timestamp	**CoT** — start/stale window (triplicate → tactical)
Source quality	`how` (e.g. `m-g`)	(none)	**CoT**
Battery / power status	`<status battery=…>`; AC state	Low battery (302), AC loss (301)	**CoT** `<status>` for telemetry value; Contact ID codes carried in detail for alarm-panel consumers (see note)
Device fault / trouble	(weak)	300-series (301 AC, 302 batt, 373 fire trouble, 380 sensor trouble)	**Contact ID** — alarm-native, richer (duplicate → alarm)
Tamper	`<status>` / detection	137 / 383 (tamper)	**Contact ID** term, CoT transport (alarm-native concept)
Detection event (heat/flame/acoustic)	`d-` sensor detection branch (`d-a` acoustic, `d-m` motion)	*(not rendered — alarm-class)*	**CoT** — `d-` sensor detection taxonomy (triplicate → tactical)
Suppression zone active	`t-`/status event; area condition	*(not rendered — alarm-class)*	**CoT** event + OSF detail sub-schema
Restoral / return to normal	new `<event>` supersedes stale	Qualifier (restoral vs. new)	**CoT** — supersede-by-freshness (triplicate → tactical)
Area condition / reservation	`r-` reservation/restriction branch	(none)	**CoT** — `r-` for wetted-zone / green-island area notices

Inbound command verbs — no external adoption (closed enum):

OSF command	Nearest external analog	Disposition
ARM / DISARM	CoT tasking `t-` (loose)	OSF-native; not adopted
WET / STOP (zone)	CoT `t-` tasking (loose)	OSF-native; domain-unique
SET CLASS (wetting class)	(none)	OSF-native
HOLD / YIELD	(none)	OSF-native
TEST / SILENCE	Contact ID test (602)	OSF-native command; publishes result as adopted status

Detail sub-schema: OSF status publishes as a CoT `<event>` whose `<detail>` carries an OSF sub-schema (the pattern CoT is designed for — an application-specific `<detail>` child under a well-known base). Where a downstream consumer is an alarm-monitoring system rather than a TAK picture, the same status renders to a Contact ID / DC-09 message using the alarm-native codes above. Both are outbound translations at the router; neither is ever accepted inbound as a command. The full CoT type-code assignments (the exact `d-`, `t-`, and `r-` leaf strings OSF emits) and the OSF↔CoT and OSF↔Contact ID mapping tables are drafted with the reference implementation and maintained under charter governance, versioned with the vocabulary itself.